

Solved Problems (IPv4 Functions)

Problem#1

Consider the network segment shown in Fig. 1. Suppose that R1 receives a packet of size 1500 Bytes with 26 Byte header. It has to be sent through R2 and R3. The R1-R2 link supports the maximum transmission unit (MTU) of 450 Bytes with a header size of 26 Bytes.

- Why is fragmentation needed in the R1-R2 link?
- Determine the data size.
- Determine the number of fragments that need to be created from the received packet in R1.
- Determine the range of byte numbers of each fragment the initial byte number of the packet is 0.
- Determine the offset number of each fragment.
- Determine the MF flag value of each fragment.
- What value of the DF field enables this fragmentation, and why?



Fig. 1

- Since the MTU (450 B) of the R1-R2 link is smaller than the packet size (1500 B), we have to divide the packet into packets of size smaller than 450 B.
- $Packet\ size = Data\ size + Header\ size$
 $1500 = Data\ size + 26$
 $Data\ size = 1500 - 26 = 1474.$
- The MTU of R1-R2 link is 450 B. Of them, 26 Bytes are packet header. Hence, maximum allowable data per fragment via the R1-R2 link is $450 - 26 = 424$ Bytes.

$$No.\ of\ fragments = \left\lceil \frac{Packet's\ Data\ size}{Fragment's\ data\ size} \right\rceil = \left\lceil \frac{1474}{424} \right\rceil = \lceil 3.48 \rceil = 4$$

Thus, breaking the 1500 bytes packet, we have to create 4 packets.

d) The byte ranges are given below:

Fragment no.	Byte number range	No. of bytes
1	0-423	424
2	424-847	424
3	848-1271	424
4	1272-1473	202

e) The formula for determining the offset of a fragment is given by

$$\text{Offset no.} = (\text{fragment no.} - 1) \times \frac{\text{Data size per fragment}}{8}$$

Offset no. of the fragment 1=

$$(1 - 1) \times \frac{424}{8} = 0 \times 53 = 0$$

Offset no. of the fragment 2=

$$(2 - 1) \times \frac{424}{8} = 1 \times 53 = 53$$

Offset no. of the fragment 3=

$$(3 - 1) \times \frac{424}{8} = 2 \times 53 = 106$$

Offset no. of the fragment 4=

$$(4 - 1) \times \frac{424}{8} = 3 \times 53 = 159$$

f) MF field indicates 'More fragment'.

MF=1 indicates that there are more fragments of the packet.

MF=0 indicates that this is the last fragment and hence there are no more fragments.

Fragment no.	MF	Explanation
1	1	This is not the last fragment. More fragments to come.
2	1	This is not the last fragment. More fragments to come.
3	1	This is not the last fragment. More fragments to come.
4	0	This is the last fragment

g) DF means “Don’t Fragment”.

If the DF field of a packet is 1. The packet cannot be fragmented (divided).

In this problem, DF field has 0 value. This is why, we are able to fragment the packet.

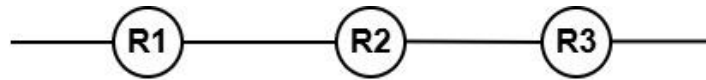


Fig. 2

Problem# 2

A router R2 receives several fragments from the router R1, with each fragment's header size is 40 bytes. Assume that the original packet has the same size header. Of them, the 7th and 11th fragments' MF fields are 1 and 0, respectively. The fragment sizes are 920 and 336 bytes, respectively.

- Determine the size of the MTU of the R2-R3 link.
- What is the size of the 10th fragment?
- Determine the size of the original packet which was fragmented by R1.
- What is the offset number of the 11th fragment?

Solution:

- The 7th fragment has 920 bytes. Since the MF field of the 7th fragment is 1, it must be a complete fragment. Hence, the MTU of the R1-R2 link is 920 bytes.
- Since the MF field of 11th fragment is 0, this is the last fragment, and only this fragment can be incomplete (i.e. less than the MTU). Hence, all previous fragments are complete fragment. Thus, similar to the 7th segment (which is also a complete fragment), this 10th fragment has the fragment size of 920 bytes.
- So far, we have learned that there are 11 fragments, of which 10 fragments are complete, and the last one is an incomplete fragment (because the size of the last fragment, 336 bytes, is less than that of a complete fragment, 920 bytes).

Data size of a complete fragment: $920 - 40 = 880$ bytes.

Data size of an incomplete fragment: $336 - 40 = 296$ bytes.

Total data size of the original packet is

$$\begin{aligned}
 &= \text{No. of incomplete fragments} \times \text{size of each complete fragment} \\
 &\quad + \text{size of the last fragment} \\
 &= 10 \times 880 + 296 = 9096 \text{ bytes}
 \end{aligned}$$

$$\begin{aligned}
 \text{Size of the original packet} &= \text{data size of the packet} + \text{header size of the packet} \\
 &= 9096 + 40 = 9136 \text{ byte}
 \end{aligned}$$

$$\begin{aligned}
 \text{d) Offset no.} &= (\text{fragment no.} - 1) \times \frac{\text{Data size per fragment}}{8} \\
 &= (11 - 1) \times \frac{880}{8} = 1100
 \end{aligned}$$

Problem# 3

An IPv4 packet is generated from a host in a university network and must pass through multiple routers to reach its destination. The packet encounters a link with a smaller MTU than the packet size. The IPv4 header of the packet contains the following important fields:

- Total Length: 1500 bytes
- Header Length (IHL): 20 bytes
- Flags: DF = 1, MF = 0
- Fragment Offset: 0

How many fragments can be created from this IPv4 packet at the router? Explain your answer.

Solution: The DF field is 1. This indicates that the fragmentation is not allowed. Since the outgoing link does not support 1500 bytes of MTU and fragmentation is not allowed, the packet will be dropped by the router and an ICMP error message will be sent to the source.

Problem#4

A router has received 8 fragments, with the header size of each fragment is 32 bytes. The last fragment's offset number is 1050 and it contains 328 bytes.

- (a) What is the offset number of the 4th fragment?
- (b) How much data has been sent in total?
- (c) How many additional bytes were needed to be sent due to the fragmentation compared to the original packet?

Solution:

a) $Offset\ no. = (fragment\ no. - 1) \times \frac{Data\ size\ per\ fragment}{8}$

$$1050 = (8 - 1) \times \frac{Data\ size\ per\ fragment}{8}$$

Data size per fragment=1200 bytes

Hence, the 4th fragment's offset number is= $(4 - 1) \times \frac{1200}{8} = 450$

- b) Data size in the last fragment: $328 - 32 = 296$ bytes.

$$Total\ data\ transmission = 7 \times 1200 + 296 = 8696\ bytes$$

- c) The header of the fragments are not explicitly mention. However, it is usually equal to that of the original packet header, we can assume that the fragment header size is also 32 bytes.

The original packet had 8696 data plus 32 bytes of header. Thus, it had $8696 + 32 = 8728$ bytes.

After fragmentation, the total data size remains the same. However, each of the 8 segments has 32 bytes of header. Hence, the total number of bytes transmitted after the fragmentation is $8696 + 8 \times 32 = 8952$.

Additional bytes needed to be sent= $8952 - 8728 = 224$.

Problem# 5

A research lab is transmitting a large IPv4 datagram containing experimental sensor data to a remote server. The total packet size is 9,064 bytes, which includes a 56-byte IPv4 header. While traversing the network, the packet reaches a router connected to a legacy link that supports a maximum transmission unit (MTU) of 1,000 bytes per frame. The router must fragment the packet before forwarding it through this link.

- Determine how many fragments will be created by the router.
- Determine the size of the last fragment (including header).
- Determine the fragment offset value of the 5th fragment, assuming that the byte number of the first byte in the packet is 16.
- Explain why the MF (More Fragments) bit is set to 0 for the last fragment.

Solution:

- a) Data size in the packet: $9064 - 56 = 9008$ bytes.

Data size in each fragment: $1000 - 56 = 944$ bytes.

No. of fragments: $\left\lceil \frac{9008}{944} \right\rceil = 10$

- b) Total data in the datagram: 9008 bytes.

First 9 fragments carry $944 \times 9 = 8496$ bytes.

The last fragment carries $9008 - 8496 = 512$ bytes of data.

Last fragment's size: data size + header size = $512 + 56 = 568$ bytes

- c) If the sequence number of the first byte does not start from 0 (as in this problem), the offset number can be computed as

$$\text{Offset no.} = (\text{fragment no.} - 1) \times \frac{\text{Data size per fragment}}{8} + \frac{\text{First byte's no.}}{8}$$

$$\text{Offset no.} = (5 - 1) \times \frac{944}{8} + \frac{16}{8} = 474$$

- d) To indicate to inform the receiver that the entire payload for that specific Datagram ID has been transmitted. It allows the receiver to calculate the total length of the original datagram using the last fragment's offset and length.

Problem# 7

An attacker, with the IP address 2.2.2.3/24, has compromised a remote device with IP address 3.3.3.4/24. The attacker's goal is to launch a Smurf attack against a victim server with IP address 5.5.5.250/24. To perform the attack, the attacker instructs the compromised device to send an ICMP Echo Request to the broadcast address of another network, 4.4.4.0/24. What should be the source IP address and the destination IP address in the crafted ICMP Echo Request packet used to perform this Smurf attack?

Solution:

Source address: 5.5.5.250 (using IP spoofing from 3.3.3.4)

Destination address: 4.4.4.255